

PHYS-467 Assignment 1

Solution

November 18, 2022

1 Theory Part

Answer the following questions by writing no more than 5 lines of text per question. Submit a single pdf file containing your answers on the assignment page on moodle. In the following d will be the dimension of the data and n will be the size of the training set.

1. (6 P.) You are training a ridge regression model with zero regularization. How does the training loss behave for $n < d$? Explain your answer.
Let $X \in \mathbb{R}^{n \times d}$ be the training data matrix. If at least n columns of X are linearly independent (which is the case when X_{ij} is random) and $n < d$, then zero training loss is achievable, independently of y . Meaning that the linear predictor can fit exactly each point. More generally one can still obtain zero training loss provided that y belongs to the span of the columns of X , even if these are not independent. As an example consider $X = \mathbf{1}_{n \times d}$ (the matrix of all ones) and $y = (1, 1, \dots, 1)$.
2. (7 P.) Let $X \in \mathbb{R}^{n \times d}$ be the training data matrix and $y \in \mathbb{R}^n$ be the labels. Under what condition on X and y is ridge regression with zero regularization able to exactly fit the training points?
 $\|y - Xw\|^2 = 0$ if w solves the linear system $y = Xw$. For this to be solvable y should belong to the span of the columns of X . If for example X_{ij} are randomly sampled and the labels y are noisy, then zero training loss is achievable iff $n \leq d$, since $\text{rank}(X) \leq \min(n, d)$.
3. (8 P.) You train ridge regression several times with fixed n and increasing regularization λ . How do you expect the training error to depend on λ ? Express the training error at $\lambda = \infty$ in terms of the labels of the training set $y = (y_1, y_2, \dots, y_n)$. Explain your reasoning
As λ increases more importance is put on minimizing $\|w\|^2$ compared to the training loss $\frac{1}{n} \|y - Xw\|^2$, hence the training loss will increase. When $\lambda \rightarrow \infty$ the fitted weights will be zero, hence the training loss gives $\frac{1}{n} \sum_{i=1}^n y_i^2$.